Lower bound reasoning with eHL

ANONYMOUS AUTHOR(S)

Notation. For a predicate P we use two indicator-style operators, one tailored to each direction of reasoning:

$$[P] \;=\; \begin{cases} 1 & \text{if } P \\ 0 & \text{otherwise} \end{cases} \qquad \text{(used for lower bounds)}$$

$$\langle P \rangle \;=\; \begin{cases} 1 & \text{if } P \\ \infty & \text{otherwise} \end{cases} \qquad \text{(used for upper bounds)}$$

Given a (state-indexed) expectation r, the expressions $[P] \cdot r$ and $\langle P \rangle \cdot r$ denote the corresponding pointwise products: $[P] \cdot r$ agrees with r on states satisfying P and is zero elsewhere, while $\langle P \rangle \cdot r$ agrees with r on states satisfying P and is ∞ elsewhere.

The witness sequence $\{r_i\}_{i \in \mathbb{N}}$. The rule of Figure 2 uses a sequence $\{r_i\}_{i \geq 0}$ of non-negative expectations as a termination witness: intuitively, r_i is a (user-chosen) upper bound on how much of the post-expectation mass of f is still trapped inside the loop after i body executions. Premise 2 ties r_i to f at $i = 0$ by requiring $r_0 \geq f$; Premise 3 propagates the bound across one iteration of the body; and Premise 4 forces the bound — and thus the trapped mass — to vanish in the limit.

2018. ACM 2475-1421/2018/1-ART1
https://doi.org/

$$\frac{}{\{\, P \mid f \,\}_{\bowtie}\ \texttt{skip}\ \{\, P \mid f \,\}}\ [\textsc{skip}]$$

$$\frac{}{\{\, P[\mathsf{x}/\mathsf{E}] \mid f[\mathsf{x}/\mathsf{E}] \,\}_{\bowtie}\ \mathsf{x} \leftarrow \mathsf{E}\ \{\, P \mid f \,\}}\ [\textsc{assign}]$$

$$\frac{}{\{\, \forall v \in \mathsf{supp}(\mathsf{S}),\ P[\mathsf{x}/v] \mid \mathbb{E}_{\mathsf{S}}[\lambda v.\, f[\mathsf{x}/v]] \,\}_{\bowtie}\ \mathsf{x} \xleftarrow{\$} \mathsf{S}\ \{\, P \mid f \,\}}\ [\textsc{sample}]$$

$$\frac{\{\, P \mid f \,\}_{\bowtie}\ \mathsf{C}\ \{\, R \mid h \,\} \quad \{\, R \mid h \,\}_{\bowtie}\ \mathsf{D}\ \{\, Q \mid g \,\}}{\{\, P \mid f \,\}_{\bowtie}\ \mathsf{C};\ \mathsf{D}\ \{\, Q \mid g \,\}}\ [\textsc{seq}]$$

$$\frac{\{\, \mathsf{B} \wedge P \mid f \,\}_{\bowtie}\ \mathsf{C}\ \{\, Q \mid g \,\} \quad \{\, \neg\mathsf{B} \wedge P \mid f \,\}_{\bowtie}\ \mathsf{D}\ \{\, Q \mid g \,\}}{\{\, P \mid f \,\}_{\bowtie}\ \texttt{if}\ \mathsf{B}\ \texttt{then}\ \mathsf{C}\ \texttt{else}\ \mathsf{D}\ \{\, Q \mid g \,\}}\ [\textsc{if}]$$

$$\frac{\mathsf{B} \wedge P \vDash f_0 \bowtie 0 \quad \forall k.\, \{\, \mathsf{B} \wedge P \mid f_{k+1} \,\}_{\bowtie}\ \mathsf{C}\ \{\, P \mid f_k \,\} \quad \neg\mathsf{B} \wedge P \vDash f_k \bowtie g}{\{\, P \mid \mathsf{lub} f_k \,\}_{\bowtie}\ \texttt{while}\ \mathsf{B}\ \texttt{do}\ \mathsf{C}\ \{\, \neg\mathsf{B} \wedge P \mid g \,\}}\ [\textsc{while}]$$

Fig. 1. eHL for lower and upper-bounds. Here $\bowtie \in \{\leq, \geq\}$.

$$\frac{\{\, \mathsf{B} \wedge P \mid f \,\}_{\leq}\ \mathsf{C}\ \{\, P \mid f \,\} \quad \mathsf{B} \wedge P \vDash r_0 \geq f \quad \forall i \geq 0.\, \{\, \mathsf{B} \wedge P \mid r_{i+1} \,\}_{\geq}\ \mathsf{C}\ \{\, \top \mid [P] \cdot r_i \,\} \quad \mathsf{B} \wedge P \vDash \lim_{i \to \infty} r_i = 0}{\{\, P \mid f \,\}_{\leq}\ \texttt{while}\ \mathsf{B}\ \texttt{do}\ \mathsf{C}\ \{\, \neg\mathsf{B} \wedge P \mid f \,\}}\ [\textsc{w}$$

Fig. 2. The While rule for eHL lower bounds. The rule is specific to the lower-bound judgment: Premise 1 is a lower-bound triple on the body, while Premise 3 is an *upper*-bound triple used purely to bound the in-loop mass. The sequence $\{r_i\}_{i \geq 0}$ consists of non-negative expectations $r_i : \mathsf{State} \to \mathbb{R}^+$.